# A Compact NAFNet U-Net for Blind, General-Purpose Image Restoration

Mini Competition: A Small Generative Model for Image Denoising — Technical Report

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## 1 Summary

We submit a 6.5 M-parameter, **4.45 GFLOPs** (fvcore, input $1\times3\times256\times256$) convolutional restoration network that maps one corrupted $256\times256\times3$ image to a restored $256\times256\times3$ `uint8` image. The model is *blind*: it receives no information about the type or strength of the corruption. Generalisation comes from training on an effectively infinite stream of randomly composed degradation chains (13 noise/blur/compression families with random order, severity and mixing) over natural, medical and low-light images. On our held-out validation set of 119 images, it raises PSNR from 21.5 to **30.6 dB**, SSIM to **0.830** and lowers DISTS to **0.179**.

**Generative formulation.** The network is a conditional generative model of the clean image given the corrupted one, $\hat{x} = G_\theta(y) \approx E[x \mid y]$, trained on synthetic pairs drawn from a hand-designed, randomised forward corruption process $y \sim q(y \mid x)$. We chose this single-step conditional generator over iterative diffusion sampling deliberately: FLOPs are ranked equally with quality, and one forward pass of our network costs roughly what a single denoising step of a diffusion U-Net costs. The perceptual (DISTS) fine-tuning stage moves the output away from the pure posterior mean towards sharper, more realistic texture.

## 2 Architecture

A 4-level U-Net built from **NAFBlocks** (Chen et al., *Simple Baselines for Image Restoration*, ECCV 2022). Each block: LayerNorm → $1\times1$ conv → $3\times3$ depth-wise conv → SimpleGate ($x_1 \odot x_2$) → simplified channel attention → $1\times1$ conv, then a gated FFN; both branches use learned residual scales initialised to zero. NAFBlocks contain no nonlinear activation functions and give state-of-the-art quality per FLOP on denoising and deblurring. Down-sampling uses stride-2 $2\times2$ convs, up-sampling uses $1\times1$ conv + PixelShuffle, and encoder features are added (not concatenated) to the decoder. A **global residual** makes the network predict a correction to the input, so clean regions are preserved. Most compute sits at $1/8$ and $1/16$ resolution, which keeps FLOPs low while giving the large receptive field that motion and defocus blur require. Only conv, element-wise and mean ops are used, so fvcore and ptflops count the model completely.

**Table 1:** Submitted configuration (`configs/small.yaml`).

| Width | Encoder blocks (levels 1–4) | Middle | Decoder blocks | Params / FLOPs @$256^2$ |
|---|---|---|---|---|
| 24 (24→384 ch.) | 1, 1, 2, 4 | 4 | 1, 1, 1, 1 | 6.53 M / 4.45 GFLOPs (fvcore), 4.52 GMACs (ptflops) |

## 3 Training data

All sources are public research or Creative Commons datasets. Large images are cut into 384 px tiles at scales 1, $\frac{1}{2}$ and $\frac{1}{4}$ (the down-scaled tiles imitate whole scenes resized to $256^2$); medical images are resized to 256 px. Training samples domains with weights natural 0.6 / medical 0.3 / low-light 0.1 (45,340 source tiles in total).

- **Natural**: DIV2K train HR (800 photos, images 0001–0740 used for training).
- **Medical**: MedMNIST+ 224 px (PneumoniaMNIST chest X-ray, OrganCMNIST CT, BreastMNIST ultrasound, RetinaMNIST fundus; CC BY 4.0) and Brain-Tumor-MRI T1/T2 slices (MIT).
- **Low-light**: LOL dataset normal-light images, plus on-the-fly darkening of any image (below).

## 4 Degradations used during training

Corruptions are synthesised *on the fly*, so no corrupted image is ever seen twice. For each patch a random chain is applied in physical image-formation order:

1. **Blur** ($p$=0.45): anisotropic **Gaussian** ($\sigma$ 0.3–3.5), linear or curved **motion** blur (3–25 px), **defocus** disk (radius 1–6).
2. **Chromatic aberration** ($p$=0.2): lateral R/B scaling and shift, optional single-channel blur.
3. **Resolution loss** ($p$=0.12): down/up-resizing by $\times$1.3–4 with random interpolation.
4. **Noise** ($p$=0.8; a second noise is mixed in with $p$=0.2): **Gaussian** ($\sigma$ 1–55, colour or luminance), **Poisson** (peak 5–500), **speckle**, **salt-and-pepper** (0.2–10%), Rician (MRI), uniform, spatially correlated (film grain / demosaicing), row/column banding.
5. **JPEG** quality 5–95 ($p$=0.35), bit-depth quantisation ($p$=0.05), late impulse noise ($p$=0.06).
6. Rounding to the `uint8` grid, matching the competition inputs.

About 3% of samples are left clean so the model learns not to over-smooth clean inputs. Content augmentation: flips/90° rotations, grayscale conversion ($p$=0.08) and **low-light simulation** ($p$=0.15: the clean target is darkened with gamma 1–2.2 and gain 0.08–0.6 before corruption). The families beyond the brief's list (Rician, banding, correlated noise, resizing) and the random combinations are intended to cover the undisclosed held-out corruptions.

## 5 Training objective and schedule

$$\mathcal{L} = \underbrace{\mathcal{L}_{\text{Charbonnier}}}_{\text{PSNR}} + 0.2\,\underbrace{(1-\text{SSIM})}_{\text{SSIM}} + 0.05\,\underbrace{\|\mathcal{F}(\hat{x}) - \mathcal{F}(x)\|_1}_{\text{sharpness after blur}} + \lambda_{\text{D}}\,\underbrace{\text{DISTS}(\hat{x}, x)}_{\text{perceptual}}, \qquad \lambda_{\text{D}} \in \{0, 0.05\}$$

Each term targets one evaluation axis. AdamW ($\beta$=(0.9, 0.9)), learning rate $10^{-3}$ with 2k-iteration warm-up and cosine decay to $10^{-6}$, gradient clipping 0.5, bf16 autocast, and an **EMA** of the weights (decay 0.999); the EMA weights are submitted. Progressive training for 200k iterations on one NVIDIA A30: iterations 0–150k with 128 px patches, batch 32; 150k–185k with 256 px patches (the true test size), batch 16; 185k–200k at 256 px with the DISTS term switched on ($\lambda_D$=0.05). The checkpoint with the best balanced validation score PSNR/40 + SSIM − DISTS is kept (here the final one).

## 6 Results

**Validation set.** 119 image pairs never used in training: 60 DIV2K photos (0741–0800), 32 MedMNIST *test* images, 12 Brain-MRI *test* images and 15 darkened LOL-eval images, each with its own random degradation chain (fixed seed). Metrics follow the competition protocol (`skimage` PSNR/SSIM and official DISTS on `uint8` $256^2$ images).

**Table 2:** Held-out validation set (final EMA weights).

| Domain | $n$ | $\text{PSNR}_{\text{in}}$ | PSNR↑ | SSIM↑ | DISTS↓ |
|---|---|---|---|---|---|
| Natural | 60 | 19.69 | 25.96 | 0.764 | 0.205 |
| Medical | 44 | 22.10 | 34.08 | 0.891 | 0.136 |
| Low-light | 15 | 27.26 | 38.86 | 0.915 | 0.203 |
| **All** | 119 | 21.53 | **30.59** | **0.830** | **0.179** |

**Table 3:** Single corruptions written independently of the training pipeline (mean of 4 validation images; PSNR dB / SSIM, input → output).

| Corruption | PSNR | SSIM |
|---|---|---|
| Gaussian $\sigma$=25 | 21.0 → 31.7 | 0.42 → 0.91 |
| Gaussian $\sigma$=50 | 15.5 → 28.6 | 0.23 → 0.83 |
| Poisson (peak 20) | 19.8 → 31.3 | 0.46 → 0.90 |
| Salt & pepper 5% | 17.6 → 40.7 | 0.46 → 0.99 |
| Speckle $\sigma$=0.2 | 25.3 → 34.3 | 0.74 → 0.95 |
| Gaussian blur $\sigma$=2 | 26.7 → 30.9 | 0.74 → 0.90 |
| Motion blur 15 px | 25.2 → 27.7 | 0.68 → 0.80 |
| JPEG $q$=10 | 26.9 → 27.9 | 0.78 → 0.83 |
| Chromatic aberration | 22.8 → 35.3 | 0.70 → 0.95 |
| Blur + noise + JPEG | 25.0 → 27.5 | 0.61 → 0.80 |
| Clean input (identity) | — → 53.5 | 1.00 → 1.00 |

The DISTS fine-tune (185k→200k) cut validation DISTS from 0.198 to 0.179 (−10%) for a 0.04 dB PSNR and 0.001 SSIM cost, the best trade under per-axis ranking. Clean inputs pass through almost unchanged (53.5 dB), so the model does not hallucinate or over-smooth.

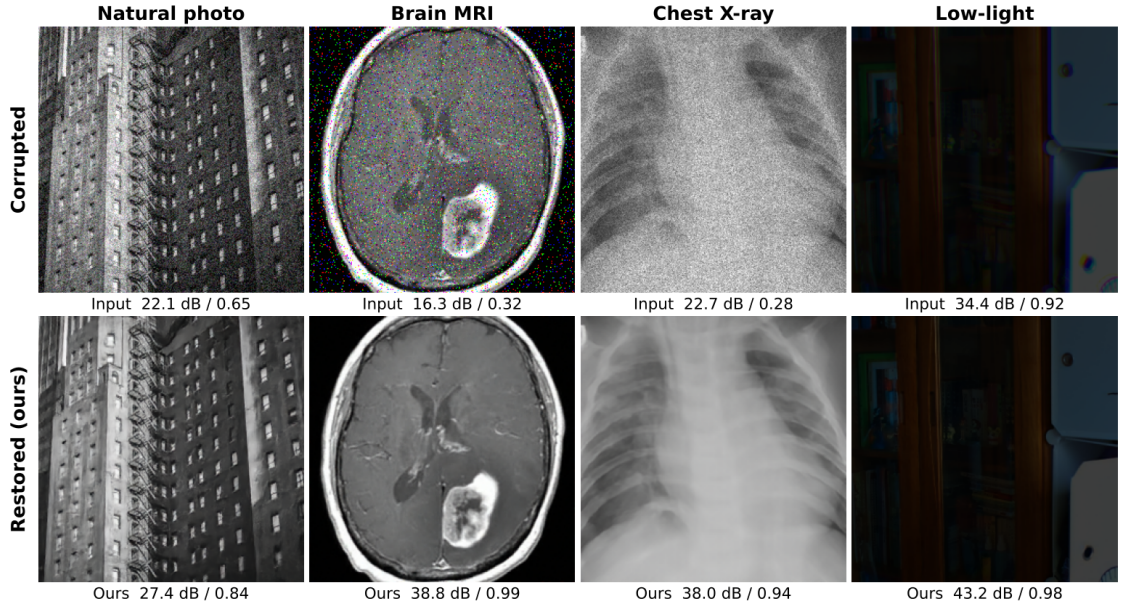



**Figure 1:** Validation examples across the three content domains (labels: PSNR / SSIM against the clean ground truth).

## 7 Limitations

- The validation corruptions come from the same family of generators as training, so the held-out score will likely be lower than 30.6 dB; Table 3 (independent corruptions) is the more honest estimate.
- Strong motion blur and very low-quality JPEG are the hardest cases (+2.5 and +1.0 dB). At high noise, fine natural texture is smoothed (e.g. animal fur).
- Global photometric changes outside the training families (haze, contrast loss) are not corrected.

## 8 Submission contents

`weights/model.pth` (PyTorch {`"state_dict"`,`"arch"`}, fp32) and `weights/model.onnx` (opset 17, verified against PyTorch, max. abs. diff. $4\times10^{-7}$). Inference: `python inference.py --input <img|dir> --output <dir>`; output is always `uint8` in $[0, 255]$. FLOPs: `python count_flops.py --weights weights/model.pth`. Training, data and Slurm scripts are included.